

VAT Computation System

Terrence Muromba · Fiscal Management graduate · Python, rules-based automation and AI-assisted development

Where the idea came from

INTERVIEW QUESTION

A client brings their data and needs their VAT computation, or other ZIMRA work, prepared, but there is very little time left before the deadline. How do you meet it?

MY ANSWER

I would use my Python and AI skills to automate the repetitive work, then check the results for accuracy.

After the interview I built a working system to show that answer in practice. A business uploads its bank statements, mobile money exports or sales and purchase lists, and gets a VAT computation it can review, with every figure traced back to its source rows.

The method behind the answer

STEP	WHAT I DO	HOW THE SYSTEM DOES IT
1. Triage	Confirm the deadline, the cost of missing it, and what data is missing.	Shows the due date (25th after the period ends), days left, the late penalty estimate, overdue returns, and months with no sales or purchases.
2. Automate	Remove the manual capture and sorting.	Reads Excel, CSV, bank and EcoCash files; classifies each transaction from rules and the regulation item lists; computes VAT per transaction at the rate in force on its date.
3. Judge the exceptions	Spend professional time only where judgement is needed.	Asks plain-language questions about what it can't decide; lists missing tax invoices, duplicates, VAT mismatches and missing exchange rates.
4. Verify, then file	Check accuracy before anything reaches ZIMRA.	Every return figure links to its transactions; finalising locks the period; every change is in the audit log.

How accuracy is protected

The tax is calculated by fixed rules, so the same data always gives the same figures, and each classification records its reason and source document. AI is planned as an assistant that suggests and explains; it never calculates the tax, and a person confirms every suggestion. The system has 112 automated tests, including worked examples checked by hand.

The system on demo data

The demo company, **Sample Haulage (Pvt) Ltd**, is a fictional transport business filing monthly (Category C). Its data is four files of the kind clients actually bring.

DEMO FILE	CONTENTS
Sales ledger (Excel)	9 September 2026 invoices in USD and ZiG: local freight, cross-border freight and a credit note
Purchases (CSV)	10 supplier invoices: diesel, tyres, a truck, import VAT, entertainment and rent
Bank statement (CSV)	An August 2026 statement with salaries, NSSA, a ZIMRA payment and bank charges
EcoCash export (CSV)	August 2026 mobile money receipts and payments

SEPTEMBER 2026 RESULT

VAT to pay in ZiG 3,616.66	ZIMRA owes in USD 13,895.35	Due date 25 Oct 2026	Questions for the user 18 groups
--------------------------------------	---------------------------------------	--------------------------------	--

The USD refund comes mainly from the VAT on a truck purchase. It can't reduce the ZiG payment, because each currency is settled separately (s38(4)).

T-Tech VATBusinessesReference listsClassification rulesVAT ratesExchange rates

Demo User · adminSign out

Businesses /
Sample Haulage (Pvt) Ltd
VAT 10099999 · BP 200123456 · Category C: monthly · reporting in USD

OverviewUploadsQuestionsTransactionsRulesAudit logSettings

The August 2026 return was due on 25 September 2026 and hasn't been filed.
Late returns cost US\$30 a day, and unpaid VAT attracts a penalty equal to the tax plus interest. File it now; if you already filed it in TaRMS, open it and click Finalise so the system knows.

VAT return for September 2026 · 21 days left
You will pay about ZiG 3,616.66 to ZIMRA by Sunday 25 October 2026.

To pay in ZiG	ZIMRA owes you in USD	Due by
3,616.66	13,895.35	25 Oct 2026

Pay each amount in its own currency. VAT received in US dollars must be paid in US dollars; paying in the wrong currency costs double the tax.

The USD 13,895.35 ZIMRA owes you can't be used to reduce the ZiG 3,616.66 you pay, because they are in different currencies.

2 things need your attention first, so the amount may change.

- 14 classification(s) need review
- 1 input tax claim(s) lack an invoice number or supplier VAT number

Answer a few questions

The overview answers “how much, in which currency, and by when” in plain words, and warns that the August return is overdue.

Based on the VAT Act [Chapter 23:12] as updated to 1 December 2024 and the 2026 National Budget Statement. Budget measures take effect once enacted in a Finance Act. All names, numbers and exchange rates in the demo are fictional.

Businesses /

Sample Haulage (Pvt) Ltd

VAT 10099999 · BP 200123456 · Category C: monthly · reporting in USD

Overview Uploads **Questions** Transactions Rules Audit log Settings**Questions about your transactions**

The system couldn't tell how VAT applies to these transactions from their descriptions. Answer in your own words below; you don't need to know the VAT categories. Similar transactions are grouped, so one answer covers them all. 21 transactions in 18 groups.

Money received · 3 transactions · 02 Sep 2026 to 29 Sep 2026

USD 6,121.50**Greenfield Millers**[See transactions](#)

Local freight Harare-Bulawayo · Local freight Harare-Gweru · Local freight Harare-Masvingo

What was this money for?

- ☐ **I sold goods or services in Zimbabwe**
Most sales. VAT is charged at the standard rate.
- ☐ **Goods exported, or transporting goods to or from another country**
No VAT is charged (0%), but keep the export documents or other proof ZIMRA accepts.
- ☐ **A service for a customer based outside Zimbabwe**
No VAT is charged (0%) if the customer is outside Zimbabwe. Keep proof, such as the contract.
- ☐ **Rent for a home, school fees, medical care, bus or train fares, or interest**
These are exempt: no VAT is charged, and VAT on related costs can't be claimed.
- ☐ **Not a sale: a loan, money from the owners, a refund, a grant, or a transfer between my own accounts**
This isn't a supply, so it doesn't go on the VAT return.

☒ Treat future transactions with Greenfield Millers the same way**Save answer**

Transactions the rules can't classify become plain questions. One answer covers a whole customer and can be remembered for future uploads.

Fiscal knowledge built in

- VAT kept separate per currency, USD and ZiG, never set off **s38(4)**
- Blocked input tax: entertainment, club fees, cars **s16(2)**
- Mixed-use apportionment, full claim at 90% taxable **s16(1)**
- Imported services self-accounted **s13**
- Rounding in the taxpayer's favour **s71**
- Rate changes and 2026 Budget measures applied by date **s73**
- Late return and late payment penalties estimated **s39, s62**
- Registration threshold of US\$25,000 tracked **s23**

Two-minute walkthrough

1. Sign in with the demo account below.
2. Open Sample Haulage: see what is owed in each currency and the overdue August return.
3. Open Questions and answer one in plain words.
4. Open the September return, click a figure, and see the transactions and exchange rates behind it.
5. Download the Excel schedules.

Try the demo

The demo uses fictional data only. Message me on LinkedIn for the demo address, then sign in with:

EMAIL **demo@example.com**
PASSWORD **demo-password-123**

Next steps

- Load the zero-rated and exempt schedules of the VAT (General) Regulations, SI 273 of 2003.
- Run one real, anonymised client period and compare the result with a return prepared by hand.
- Add a guided filing wizard, payment tracking and WhatsApp deadline reminders.